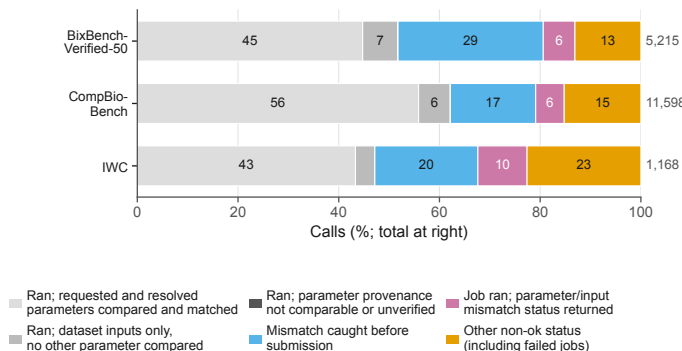
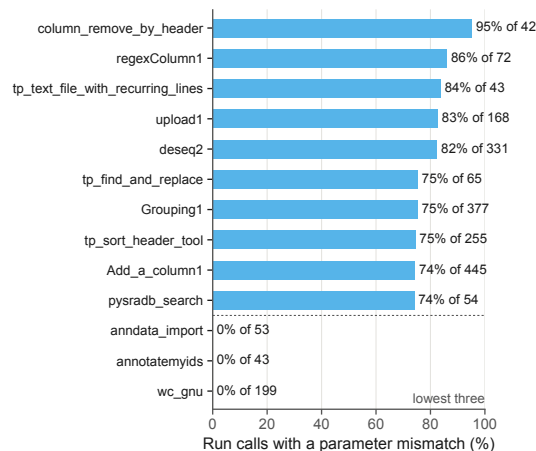


a Installed-tool run calls by parameter-check coverage and outcome**b** Parameter-mismatch rate by tool (tools with ≥40 run calls)**c** Requested, executed and interpreted analysis: five mechanisms that changed or obscured answers

Conditional parameter rebound to its default (bix-35-q1)	Requested: metric "evolutionary rate". Resolved and executed: default "total tree length"; job state ok. A resubmission that named the branch by index sent no comparable parameter, so the check compared none.	7 of 15 histories held such a job; 6 runs resubmitted after the flag; 1 answer wrong
Output semantics not declared (bix-28-q3)	Requested: one summary value per tree. The output was the variance under the metric name. Wrapper parsing is inferred; the wrapper source was not archived.	1 answer wrong: 146.3, outside the range of the four values (median ~30.5)
Underlying software version not exposed (bix-45-q1)	Search and inspect exposed wrapper versions, not PhyKIT versions. 13 runs used the wrapper and 2 used Galaxy user-defined tools; the reference encodes a retired definition.	All 15 Galaxy runs gave the same, scientifically defensible answer; all were scored incorrect
Execution environment changed tool behaviour (encode-atac-pipeline-q1)	Inputs were staged as dataset_*_dat; the pipeline detects compression by file extension, so adapter trimming was skipped without any error.	7 of 12 computed untrimmed peaks; 1 additional run copied a pilot answer
Several versions of one tool in the catalog (IWC peptide verification)	Runs that used the legacy PepQuery 1.6.2 wrapper scored lower than runs that used only PepQuery2 2.0.2 (association; search order was not controlled).	4 runs: 0.868-0.908 8 runs: 0.982-1.000

d Galaxy as a cause of wrong answers (targeted audit)